

Why an Inverse-Elliptical Potential Produces a Force-Magnitude Dip Along its Major Axis

Abstract

For the artificial potential field $U = 1/\rho$ built on an anisotropic (elliptical) pseudo-distance ρ with elongation ratio q , the force magnitude $|\nabla U|$ has a strict local *minimum* on the major axis whenever $q > \sqrt{3}$. We derive the exact threshold, the location of the off-axis ridge, and the depth of the dip in closed form, and we give the geometric reason for the constant $\sqrt{3}$. This explains the narrow low-force channel observed directly ahead of an obstacle in the DIPF force plots, and shows that the feature is a necessary consequence of elongating the field rather than a numerical artefact.

1 Setup

Place the obstacle at the origin and define the *elliptical radius*

$$\rho(x, y) = \sqrt{x^2 + \frac{y^2}{q^2}}, \quad q > 0, \quad (1)$$

where q is the elongation of the field along y . In the implementation $q = t_y/t_x$, the ratio of the two velocity-inflated time constants. The potential is the inverse of this pseudo-distance,

$$U(x, y) = \frac{1}{\rho(x, y)}. \quad (2)$$

The level sets $\{U = \text{const}\}$ are concentric ellipses with semi-axis ratio $q : 1$, the major axis lying along y . The force is $F = -\nabla U$, and the quantity plotted as the background colour field is $|F| = |\nabla U|$.

Differentiating (2),

$$\partial_x U = -x \rho^{-3}, \quad \partial_y U = -\frac{y}{q^2} \rho^{-3}, \quad (3)$$

so that

$$|\nabla U|^2 = \frac{x^2 + y^2/q^4}{\rho^6} = \frac{x^2 + y^2/q^4}{(x^2 + y^2/q^2)^3}. \quad (4)$$

2 The dip criterion

Fix a height $y = s > 0$ and scan horizontally. Introduce the *core half-width* at that height,

$$a = \frac{s}{q}, \quad (5)$$

which is the semi-minor axis of the equipotential ellipse passing through $(0, s)$. Substituting $y = s$ into (4) and changing variables to $u = x^2 \geq 0$ gives

$$g(u) := |\nabla U|^2 \Big|_{y=s} = \frac{u + a^2/q^2}{(u + a^2)^3}. \quad (6)$$

Because $|\nabla U|^2$ depends on x only through $u = x^2$, it is an even function of x and its x -derivative vanishes at $x = 0$ automatically. The substantive question is therefore not whether $x = 0$ is a critical point, but whether it is a minimum or a maximum. That is decided by the sign of g' .

Theorem 1 (Dip criterion). *Let g be as in (6). Then*

$$g'(u) = \frac{a^2 \left(1 - \frac{3}{q^2}\right) - 2u}{(u + a^2)^4}, \quad u \geq 0. \quad (7)$$

Consequently:

1. *If $q \leq \sqrt{3}$, then $g'(u) \leq 0$ for all $u \geq 0$, so $|\nabla U|$ is non-increasing in $|x|$ and the major axis carries the largest force on the scan line. There is no dip.*
2. *If $q > \sqrt{3}$, then $g'(0) > 0$, so $x = 0$ is a strict local minimum of $|\nabla U|$, and the unique maximum on the scan line lies at*

$$x^* = \frac{a}{\sqrt{2}} \sqrt{1 - \frac{3}{q^2}} = \frac{s}{q\sqrt{2}} \sqrt{1 - \frac{3}{q^2}} \xrightarrow{q \rightarrow \infty} \frac{s}{q\sqrt{2}}. \quad (8)$$

Proof. Write $g(u) = N(u)/D(u)$ with $N = u + a^2/q^2$ and $D = (u + a^2)^3$. Then $N' = 1$ and $D' = 3(u + a^2)^2$, so by the quotient rule

$$\begin{aligned} g'(u) &= \frac{(u + a^2)^3 - (u + a^2/q^2) \cdot 3(u + a^2)^2}{(u + a^2)^6} \\ &= \frac{(u + a^2) - 3(u + a^2/q^2)}{(u + a^2)^4} = \frac{a^2 \left(1 - \frac{3}{q^2}\right) - 2u}{(u + a^2)^4}, \end{aligned} \quad (9)$$

where the second line cancels one factor of $(u + a^2)^2$. This is (7).

Since $a > 0$, the denominator $(u + a^2)^4$ is strictly positive on $u \geq 0$. Hence $\text{sign } g'(u) = \text{sign}[a^2(1 - 3/q^2) - 2u]$, an affine and strictly decreasing function of u .

If $q^2 \leq 3$ the constant term $a^2(1 - 3/q^2)$ is non-positive, so the numerator is ≤ 0 throughout $u \geq 0$; thus g is non-increasing, and so is $|\nabla U| = \sqrt{g}$ (the square root is monotone). Since $u = x^2$ increases with $|x|$, claim 1 follows.

If $q^2 > 3$ the numerator is positive at $u = 0$ and decreases affinely, vanishing exactly once at

$$u^* = \frac{1}{2}a^2 \left(1 - \frac{3}{q^2}\right) > 0, \quad (10)$$

after which it is negative. Hence g strictly increases on $[0, u^*]$ and strictly decreases on $[u^*, \infty)$: $u = 0$ is a strict local minimum and u^* is the unique global maximum. Taking $x^* = \sqrt{u^*}$ gives (8). $\square$

Remark 2. *A circle ($q = 1$) can never produce the dip, and any elongation beyond $q = \sqrt{3} \approx 1.732$ must. The feature is thus not an artefact of the particular potential used, but a property of elongation itself.*

3 Depth of the dip

Theorem 3 (Contrast ratio). *For $q > \sqrt{3}$, the ratio of the ridge force to the on-axis force on any horizontal scan line is*

$$C := \frac{|\nabla U|_{\max}}{|\nabla U|_{x=0}} = \frac{2q^3}{3\sqrt{3}(q^2-1)} \xrightarrow{q \rightarrow \infty} \frac{2q}{3\sqrt{3}} \approx 0.385q. \quad (11)$$

In particular C is independent of the height s .

Proof. From (6), on the axis

$$g(0) = \frac{a^2/q^2}{a^6} = \frac{1}{q^2 a^4}. \quad (12)$$

At the ridge, substitute u^* from (10). The two brackets simplify because the $3/q^2$ terms combine:

$$u^* + \frac{a^2}{q^2} = \frac{a^2}{2} \left(1 - \frac{3}{q^2}\right) + \frac{a^2}{q^2} = \frac{a^2}{2} \left(1 - \frac{1}{q^2}\right), \quad (13)$$

$$u^* + a^2 = a^2 \left[1 + \frac{1}{2} \left(1 - \frac{3}{q^2}\right)\right] = \frac{3a^2}{2} \left(1 - \frac{1}{q^2}\right). \quad (14)$$

Writing $\beta = 1 - q^{-2}$,

$$g(u^*) = \frac{\frac{1}{2}a^2\beta}{\left(\frac{3}{2}a^2\beta\right)^3} = \frac{\frac{1}{2}a^2\beta}{\frac{27}{8}a^6\beta^3} = \frac{4}{27a^4\beta^2}. \quad (15)$$

Therefore

$$C^2 = \frac{g(u^*)}{g(0)} = \frac{4}{27a^4\beta^2} \cdot q^2 a^4 = \frac{4q^2}{27\beta^2}, \quad (16)$$

and taking the positive square root with $\beta = (q^2 - 1)/q^2$ gives

$$C = \frac{2q}{3\sqrt{3}\beta} = \frac{2q^3}{3\sqrt{3}(q^2-1)}. \quad (17)$$

Both a and s have cancelled, proving the height-independence. $\square$

Corollary 4 (Geometry of the channel). *Since C does not depend on s while $x^* \propto s$ by (8), the locus of ridge points is a pair of straight rays through the obstacle,*

$$x = \pm \kappa s, \quad \kappa = \frac{1}{q} \sqrt{\frac{1}{2} \left(1 - \frac{3}{q^2}\right)}, \quad (18)$$

enclosing a wedge of uniformly suppressed force. The dip therefore appears in a plot as a narrow channel of constant contrast opening away from the obstacle, i.e. the observed “U” shape, rather than as an isolated blob.

4 Why the constant is $\sqrt{3}$

The threshold has a direct geometric reading. Factor the gradient through the metric:

$$|\nabla U| = |U'(\rho)| \underbrace{|\nabla \rho|}_{\text{amplitude}} = \underbrace{\rho^{-2}}_{\text{metric}} |\nabla \rho|, \quad |\nabla \rho| = \frac{\sqrt{x^2 + y^2/q^4}}{\rho}. \quad (19)$$

Moving off-axis at fixed height, the two factors oppose one another:

- $|\nabla\rho|$ **increases** from $1/q$ on the major axis to 1 on the minor axis. Equipotentials are q times more widely spaced at the poles of the ellipse than at its flanks, so the force there is q times weaker. This is the elongation, and it favours an off-axis ridge.
- ρ^{-2} **decreases**, since ρ grows with $|x|$. This is the inverse-distance decay, and it favours a monotone falloff.

The exponent 3 appearing in (7) is precisely the power ρ^{-3} carried by the gradient of a $1/\rho$ potential, as seen in (3). The dip survives only when the metric gain outpaces that decay, which requires $q^2 > 3$. A different radial power law shifts the threshold accordingly; an isotropic field has no metric gain at all ($|\nabla\rho| \equiv 1$) and so never exhibits the dip.

5 Numerical verification

Evaluating $|\nabla U|$ by central differences on (2) at $s = 8$ over a fine grid in x reproduces both closed forms to four decimal places.

q	x^* predicted	x^* numerical	C predicted	C numerical
1.50	—	—	1.0000	1.0000
$\sqrt{3}$	0.0000	0.0149	1.0000	1.0000
2	1.4142	1.4142	1.0264	1.0264
3	1.5396	1.5396	1.2990	1.2990
6	0.9027	0.9027	2.3754	2.3754
12	0.4665	0.4665	4.6511	4.6511
30	0.1882	0.1883	11.5598	11.5598

At $q = \sqrt{3}$ the maximum is degenerate ($x^* = 0$) and the numerical value merely reflects the grid resolution, consistent with $C = 1$.

6 Application to the implemented field

The implemented potential carries an additional vehicle-dimension prefactor,

$$U_{\text{impl}}(x, s) = \frac{\lambda}{(1 - l/s) \rho(x, s)}, \quad (20)$$

with l the obstacle length and s the longitudinal gap. Two consequences:

- **The ridge location is unchanged.** The prefactor $(1 - l/s)^{-1}$ is constant in x along a horizontal scan, so it rescales the x -profile without shifting its stationary point. For $q = 12$ at $s = 9.5$, (8) predicts $x^* = 0.554$ against a measured 0.54 in the implementation.
- **The dip is shallower.** The factor does depend on s , so ∂_y acts on it and contributes an additional, x -dependent term to F_y that partially fills the on-axis minimum. The measured contrast is $C \approx 2.52$ against the pure-elliptical value 4.6511 — qualitatively identical, roughly half as deep.

Finally, note that the dip lives in $|\nabla U|$, not in U . The potential itself is maximal on the major axis and decreases monotonically outward, so its level sets remain nested convex ellipses; it is the level sets of the *force magnitude* that become non-convex and pinched at the poles. In control terms the major axis is a ridge of U , hence an unstable equilibrium line: a host vehicle exactly on the centreline receives a near-zero lateral command, while a small lateral offset receives a large one.